

Учреждение образования
«Брестский государственный технический университет»
Кафедра «ЭВМ и систем»

**DESIGN OF INTELLIGENT EDGE AND TIME-SERIES DATA FLOW
TEMPERATURE CONTROL SYSTEM**

TEXT PROGRAM

(на оптическом носителе CD)

БрГТУ. 41297 - 06 12 00

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Объем 2247 КБайт

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Abstract

EdgeHub-Based Closed-Loop Temperature Control System is a software and hardware prototype for automatic temperature stabilization in an edge-control environment. It combines an ESP32-based edge node, MQTT telemetry exchange, a Java Data Hub, and a web HMI into one monitoring and control solution. The edge node reads a digital temperature sensor, filters measurements, executes a PI control algorithm, and drives a PWM heater to keep the process near the configured set point. It also accepts safe parameter updates and reports status without stopping the control process. The Data Hub validates MQTT messages, tracks device state, stores normalized events in TDengine or archive files, and separates raw device traffic from application logic. The HMI exposes REST services and a React interface for live temperature, heater output, control mode, alarms, history, and parameter forms. Operators can review behavior, adjust bounded settings, send commands, and confirm acknowledgements. An AI-assisted tuning workflow analyzes recent telemetry, prepares PID recommendations, checks safety limits, publishes selected values, and compares actual behavior with the predicted response. The program demonstrates the complete path from sensor measurement to actuator response and provides a replaceable basis for future testing and integration with industrial monitoring services.

Content

README.md - file with the general source-code package description;
data-hub/ - directory with Java data-ingestion source code;
 README.md - file with Data Hub module description;
 build.gradle, settings.gradle - Gradle project and dependency configuration;
 config/application.example.properties - example Data Hub configuration file;
 src/main/java/com/edgehub/datahub/DataHubApplication.java - service startup code;
 src/main/java/com/edgehub/datahub/alarm/ - directory with alarm rule evaluation code;
 src/main/java/com/edgehub/datahub/config/ - directory with service configuration code;
 src/main/java/com/edgehub/datahub/model/ - directory with telemetry and status models;
 src/main/java/com/edgehub/datahub/mqtt/ - directory with MQTT consumer code;
 src/main/java/com/edgehub/datahub/parser/ - directory with topic and payload parser code;
 src/main/java/com/edgehub/datahub/pipeline/ - directory with message processing code;
 src/main/java/com/edgehub/datahub/storage/ - directory with TDengine and archive writers;
edge-firmware/ - directory with ESP32 firmware source code;
 README.md - file with firmware module description;
 diagram.json, wokwi.toml - Wokwi simulation project files;
 platformio.ini - PlatformIO project configuration for firmware build;
 src/app/edge_app.cpp - code file with the main edge-node workflow;
 src/app/runtime_config_store.cpp - code file for stored runtime parameters;
 src/comms/mqtt/ - directory with MQTT gateway, telemetry, ack and parameter code;
 src/config/ - directory with firmware build and pin configuration;
 src/controller/pi_controller.cpp - code file with PI temperature control;
 src/domain/model/ - directory with control state and message models;
 src/hardware/interfaces/ - directory with sensor and actuator interfaces;
 src/hardware/real/ - directory with real DS18B20 sensor and heater driver code;
 src/hardware/wokwi/ - directory with simulated sensor, heater and plant model code;
 src/sketch.ino - code file with the firmware entry point;
hmi/backend/ - directory with Python FastAPI backend source code;
 .env.example, requirements.txt, alembic.ini - backend configuration files;
 ai/ - directory with AI service scripts and supporting documentation;
 alembic/ - directory with database migration source files;
 app/api/routes/ - directory with routes for devices, history, alarms, users and operations;
 app/core/ - directory with configuration, logging and security code;
 app/db/ - directory with database session code;
 app/main.py - code file with API application startup;
 app/models/, app/schemas/ - directories with database entities and API schemas;
 app/services/ - directory with MQTT publishing, TDengine access, AI tuning and diagnostics code;
hmi/frontend/ - directory with React TypeScript HMI source code;
 .env.example, package.json, vite.config.ts, tsconfig.json - frontend configuration files;

index.html - frontend HTML entry file;

src/app/ - directory with authentication and global style code;

src/components/ - directory with layout and reusable UI component code;

src/lib/api.ts - code file with REST API client functions;

src/main.tsx - code file with frontend application startup;

src/pages/ - directory with overview, device, history, alarm, AI, user and operation pages;

src/routes/ - directory with application routing and data loading code;

src/types/ - directory with shared TypeScript type definitions.

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